# DEMONSTRATION CANDIDATE ASSESSMENT BRIEF

**Assessment Reference:** EDU-DIG-001
**Assessment Title:** Counter 4-bit
**Assessment Level:** Demonstration / Introductory
**Assessment Type:** Digital Circuit Analysis
**Analysis Software:** EDUPREP Digital Waveform Viewer
**Filename:** EDUPREP_Counter4_golden.vcd

---

# 1. Introduction

Welcome to the EDUPREP Digital Waveform Viewer demonstration exercise.

This exercise is designed to introduce you to digital waveform analysis and the investigation of abnormal digital circuit behaviour.

You are required to examine the supplied simulation data, interpret the digital signals and identify any differences between the expected and observed circuit behaviour.

# 2. Assessment Scenario

You have been provided with simulation data from a 4-bit binary counter.

The counter is designed to increment its output on each rising edge of the clock signal. It also has a synchronous clear input, which resets the counter to zero when activated.

Two simulation files are provided:

- **EDUPREP_Counter4_golden.vcd** – Simulation data representing the expected circuit behaviour.
- **EDUPREP_Counter4_faulty.vcd** – Simulation data representing the behaviour of a circuit containing an induced fault.

Your task is to compare the two simulations and determine whether the faulty circuit behaves as expected.

# 3. Circuit Description

The circuit under investigation is a 4-bit binary counter with the following signals:

| Signal | Description |
|---|---|
| `clk` | Clock input |
| `CLR` | Synchronous clear input |
| `Q[3:0]` | 4-bit counter output |

Under normal operating conditions:

- The counter starts at `0000`.
- The counter increments by one on every rising clock edge.
- The output progresses through the binary sequence from `0000` to `1111`.
- After reaching `1111`, the counter rolls over to `0000`.
- When `CLR` is HIGH at a rising clock edge, the counter resets to `0000`.

# 4. Assessment Objectives

Using the EDUPREP Digital Waveform Viewer, you are required to:

1. Load and examine the golden simulation file.
2. Identify the clock signal and determine the counter's operating sequence.
3. Examine the 4-bit output bus and interpret its binary values.
4. Load and examine the faulty simulation file.
5. Compare the faulty output sequence with the golden reference.
6. Identify any signal or signals that do not behave as expected.
7. Determine the nature of the abnormal behaviour.
8. Explain the likely effect of the fault on the operation of the counter.

# 5. Candidate Investigation

During your investigation, consider the following:

- Does the counter increment on every rising edge of the clock?
- Does the output follow the expected binary counting sequence?
- Does the output reach all possible 4-bit binary values?
- Are all four output bits changing as expected?
- Can you identify any repeated or missing binary values?
- Is the observed behaviour consistent with a correctly operating counter?

Use the waveform viewer's available navigation, zoom and measurement features where necessary to examine the timing relationships between the signals.

# 6. Candidate Findings

Record your observations and conclusions in the following areas:

## A. Golden Reference

Describe the expected counter operation and the observed output sequence.

## B. Faulty Simulation

Describe any differences between the golden and faulty simulation results.

### C. Fault Identification

Identify the signal or signals exhibiting abnormal behaviour and explain what you observed.

### D. Technical Conclusion

Explain the likely effect of the identified fault on the counter's operation.

# 7. Assessment Expectations

The candidate is expected to demonstrate the ability to:

- Navigate and interpret digital simulation waveforms.
- Understand binary counting and multi-bit signal representations.
- Recognise abnormal signal behaviour.
- Compare actual circuit behaviour against a known-good reference.
- Use waveform evidence to support a technical conclusion.

Candidates should base their findings on the supplied simulation data rather than assumptions.

# 8. Assessment Deliverable

Submit your completed investigation findings to the evaluator.

Your conclusions should clearly identify the abnormal behaviour and be supported by observations from the waveform data.

**End of Assessment Brief**

*Developing practical skills through applied technical assessment.*